

Technical Appendix (Extended Version): Complete Field Equations and Full Derivations of the Two-Mode Coupling Model (Track A)

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Abstract

This extended appendix collects, in a single self-contained reference, all the field equations used in the two-mode Jaynes–Cummings model of Track A, together with the full derivations of three key technical results: (i) the mean-field closure of the reduced dynamics, (ii) the expansion of the density matrix in the eigenbasis of the Jaynes–Cummings Hamiltonian, and (iii) the linear stability analysis of the Liouvillian superoperator. We present the Hilbert-space structure, the operator algebra, the Hamiltonian, the Lindblad dissipator, the master equation, the reduced dynamics, the control parameters, the mixing angle, the order parameters, the information-theoretic measures, and the numerical scheme. The three new sections contain step-by-step derivations intended to make the model fully reproducible and to expose the mathematical structure underlying the numerical results of the main paper.

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1 Conventions and notation

1.1 Units

Throughout this appendix we set $\hbar = 1$ and $k_B = 1$. Frequencies are measured in Hz (equivalently, in units of s^{-1}). Energy is measured in the same units as frequency. When dimensional consistency requires it, $\hbar$ is restored explicitly.

1.2 Indices

- 0: substrate sector (non-analytic, oscillator mode).
- 1: analytic sector (baryonic, qubit mode).

- N : total excitation number.
- $n \in \{0, 1, \dots, N_0 - 1\}$: Fock quantum number of the oscillator.
- $s \in \{0, 1\}$: qubit state label.
- μ, ν : Liouvillian eigenmode indices.

1.3 Operators

For any operator $\hat{O}$ we denote by

- $\hat{O}^\dagger$: Hermitian conjugate.
- $\langle \hat{O} \rangle_\rho = \text{Tr}(\rho \hat{O})$: expectation value in state ρ .
- $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$: commutator.
- $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$: anticommutator.
- $\text{vec}(\hat{O})$: column vector obtained by stacking columns of $\hat{O}$.

2 Hilbert space structure

2.1 Single-sector spaces

The total Hilbert space is the tensor product

$$\boxed{\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_1, \quad \dim \mathcal{H} = d_0 \cdot d_1.} \quad (1)$$

For the numerical implementation we truncate the oscillator to N_0 levels, so

$$d_0 = N_0, \quad d_1 = 2, \quad \dim \mathcal{H} = 2N_0 \equiv d. \quad (2)$$

2.2 Basis of the analytic sector

We use the basis

$$\mathcal{B}_1 = \{|0\rangle_1, |1\rangle_1\}, \quad (3)$$

where

- $|0\rangle_1$: analytic mode *off* (inactive).
- $|1\rangle_1$: analytic mode *on* (active).

The projectors are

$$\hat{P}_0^{(1)} = |0\rangle_1 \langle 0|, \quad \hat{P}_1^{(1)} = |1\rangle_1 \langle 1|, \quad (4)$$

satisfying

$$\hat{P}_0^{(1)} + \hat{P}_1^{(1)} = \hat{I}_1, \quad \hat{P}_i^{(1)} \hat{P}_j^{(1)} = \delta_{ij} \hat{P}_i^{(1)}. \quad (5)$$

2.3 Basis of the substrate sector

We use the Fock basis

$$\mathcal{B}_0 = \{|n\rangle_0\}_{n=0}^{N_0-1}. \quad (6)$$

The number operator is

$$\hat{n}_0 = \hat{a}^\dagger \hat{a}, \quad \hat{n}_0 |n\rangle_0 = n |n\rangle_0. \quad (7)$$

Completeness in the truncated space:

$$\sum_{n=0}^{N_0-1} |n\rangle_0 \langle n| = \hat{I}_0. \quad (8)$$

2.4 Tensor-product operators

For any operator $\hat{A}$ acting on $\mathcal{H}_0$ and any operator $\hat{B}$ acting on $\mathcal{H}_1$, we write

$$\hat{A} \otimes \hat{B} \equiv \hat{A}\hat{B}, \quad (9)$$

with the understanding that the first factor acts on the substrate and the second on the analytic sector. When no ambiguity arises, we write $\hat{A}$ for $\hat{A} \otimes \hat{I}_1$ and $\hat{B}$ for $\hat{I}_0 \otimes \hat{B}$.

3 Operator algebra

3.1 Oscillator operators

The annihilation and creation operators satisfy

$$\boxed{[\hat{a}, \hat{a}^\dagger] = \hat{I}_0, \quad [\hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}^\dagger] = 0.} \quad (10)$$

Their action on the Fock basis is

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle, \quad (11)$$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle. \quad (12)$$

The number operator satisfies

$$\hat{n}_0 = \hat{a}^\dagger \hat{a}, \quad [\hat{n}_0, \hat{a}] = -\hat{a}, \quad [\hat{n}_0, \hat{a}^\dagger] = +\hat{a}^\dagger. \quad (13)$$

In the matrix representation (truncated to N_0 levels), the annihilation operator is

$$(\hat{a})_{mn} = \sqrt{m} \delta_{m+1,n}, \quad m, n \in \{0, 1, \dots, N_0 - 1\}. \quad (14)$$

3.2 Qubit operators

The Pauli operators in the basis $\mathcal{B}_1$ are

$$\hat{\sigma}_+ = |1\rangle_1 \langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (15)$$

$$\hat{\sigma}_- = |0\rangle_1 \langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (16)$$

$$\hat{\sigma}_z = |1\rangle_1 \langle 1| - |0\rangle_1 \langle 0| = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}. \quad (17)$$

They satisfy the algebra

$$[\hat{\sigma}_+, \hat{\sigma}_-] = \hat{\sigma}_z, \quad [\hat{\sigma}_z, \hat{\sigma}_\pm] = \pm 2\hat{\sigma}_\pm. \quad (18)$$

The projector onto the active state is

$$\hat{\sigma}_+ \hat{\sigma}_- = |1\rangle_1 \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (19)$$

3.3 Total excitation operator

We define the total excitation number

$$\hat{\mathcal{N}} = \hat{a}^\dagger \hat{a} + \hat{\sigma}_+ \hat{\sigma}_- = \hat{n}_0 + \hat{\sigma}_+ \hat{\sigma}_-. \quad (20)$$

Under the Jaynes–Cummings interaction, this is conserved:

$$[\hat{\mathcal{N}}, \hat{H}_{\text{int}}] = 0. \quad (21)$$

4 Hamiltonians

4.1 Free Hamiltonian

The free Hamiltonian is

$$\hat{H}_0 = \omega_0 \hat{a}^\dagger \hat{a} + \omega_1 \hat{\sigma}_+ \hat{\sigma}_-. \quad (22)$$

Its action on the product basis is

$$\hat{H}_0 |n\rangle_0 \otimes |s\rangle_1 = (\omega_0 n + \omega_1 \delta_{s,1}) |n\rangle_0 \otimes |s\rangle_1. \quad (23)$$

4.2 Detuning

The detuning between sectors is

$$\delta \equiv \omega_1 - \omega_0. \quad (24)$$

In the numerical simulations, we use $\omega_0 = 6$ Hz and $\omega_1 = 40$ Hz, giving $\delta = 34$ Hz.

4.3 Interaction Hamiltonian

The Jaynes–Cummings interaction in the Rotating Wave Approximation (RWA) is

$$\hat{H}_{\text{int}}(t) = g(t) \left(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+ \right). \quad (25)$$

The two terms have a clear physical interpretation:

- $\hat{a}^\dagger \hat{\sigma}_-$: creates a substrate quantum and de-excites the analytic mode.
- $\hat{a} \hat{\sigma}_+$: destroys a substrate quantum and excites the analytic mode.

The RWA is valid when

$$g \ll \omega_0 + \omega_1. \quad (26)$$

4.4 Total Hamiltonian

The total Hamiltonian is

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_{\text{int}}(t). \quad (27)$$

In the product basis $|n\rangle_0 \otimes |s\rangle_1$:

$$\hat{H}(t) = \sum_{n,s} E_{n,s} |n, s\rangle \langle n, s| + g(t) \sum_{n=0}^{N_0-2} \sqrt{n+1} (|n+1, 0\rangle \langle n, 1| + |n, 1\rangle \langle n+1, 0|), \quad (28)$$

where

$$E_{n,s} = \omega_0 n + \omega_1 \delta_{s,1}. \quad (29)$$

5 Rabi oscillation and mixing angle

5.1 Excitation manifolds

For a fixed total excitation number N , the states

$$|N, 0\rangle \equiv |N\rangle_0 \otimes |0\rangle_1, \quad |N-1, 1\rangle \equiv |N-1\rangle_0 \otimes |1\rangle_1, \quad (30)$$

span a two-dimensional subspace. In this subspace, the total Hamiltonian reduces to

$$\hat{H}^{(N)} = \begin{pmatrix} \omega_0 N & g\sqrt{N} \\ g\sqrt{N} & \omega_0(N-1) + \omega_1 \end{pmatrix} = \omega_0(N-1) + \omega_1 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} \delta & g\sqrt{N} \\ g\sqrt{N} & 0 \end{pmatrix}. \quad (31)$$

Shifting the zero of energy by $\omega_0(N-1) + \omega_1$:

$$\hat{H}_{\text{shifted}}^{(N)} = \begin{pmatrix} \delta & g\sqrt{N} \\ g\sqrt{N} & 0 \end{pmatrix}. \quad (32)$$

5.2 Eigenvalues and eigenvectors

The eigenvalues of $\hat{H}_{\text{shifted}}^{(N)}$ are

$$E_{\pm}^{(N)} = \frac{\delta}{2} \pm \frac{1}{2} \sqrt{\delta^2 + 4g^2 N}. \quad (33)$$

The corresponding eigenvectors are

$$|+, N\rangle = \cos \theta_N |N, 0\rangle + \sin \theta_N |N-1, 1\rangle, \quad (34)$$

$$|-, N\rangle = -\sin \theta_N |N, 0\rangle + \cos \theta_N |N-1, 1\rangle, \quad (35)$$

with the mixing angle

$$\boxed{\theta_N = \frac{1}{2} \arctan \left(\frac{2g\sqrt{N}}{\delta} \right)}. \quad (36)$$

5.3 Rabi frequency

We define the Rabi frequency at excitation manifold N as

$$\boxed{\Omega_R^{(N)} \equiv \sqrt{\delta^2 + 4g^2 N}}. \quad (37)$$

For $N = 1$:

$$\Omega_R^{(1)} = \sqrt{\delta^2 + 4g^2} = 2\sqrt{g^2 + (\delta/2)^2}. \quad (38)$$

The corresponding Rabi period is

$$T_R = \frac{2\pi}{\Omega_R^{(1)}}. \quad (39)$$

For $g = 17$ Hz, $\delta = 34$ Hz: $\Omega_R^{(1)} = 48.08$ Hz, $T_R = 0.131$ s.

5.4 Geometric labels

The value $\theta_N = \pi/8 = 22.5^\circ$ corresponds to

$$\frac{2g\sqrt{N}}{\delta} = \tan(\pi/4) = 1 \implies g_{\text{crit}}^{(N)} = \frac{\delta}{2\sqrt{N}}. \quad (40)$$

For $N = 1$: $g_{\text{crit}}^{(1)} = \delta/2 = 17$ Hz.

Important: This value is a *geometric label*, not a dynamical transition. Sweeping g across g_{crit} does not produce a qualitative change in the dynamics.

6 DMN modulation

6.1 Activity variable

We introduce the DMN activity variable

$$A_{\text{DMN}}(t) \in [0, 1]. \quad (41)$$

6.2 Sigmoidal filter

The coupling modulation is

$$\eta(A) = \frac{1}{1 + \exp[\lambda(A - A_{\text{th}})]}, \quad (42)$$

with λ the stiffness and A_{th} the threshold.

6.3 Coupling strength

The time-dependent coupling is

$$g(t) = g_0 \eta(A_{\text{DMN}}(t)). \quad (43)$$

6.4 Stochastic dynamics

The DMN activity evolves as an Ornstein–Uhlenbeck process:

$$\tau_{\text{DMN}} \frac{dA_{\text{DMN}}}{dt} = -(A_{\text{DMN}} - A_0) + \sigma \xi(t), \quad (44)$$

with

$$\langle \xi(t) \rangle = 0, \quad \langle \xi(t) \xi(t') \rangle = 2\delta(t - t'). \quad (45)$$

The stationary distribution is Gaussian with

$$\langle A_{\text{DMN}} \rangle = A_0, \quad \text{Var}(A_{\text{DMN}}) = \frac{\sigma^2}{\tau_{\text{DMN}}}. \quad (46)$$

7 Lindblad dissipation

7.1 Master equation

The evolution of the density matrix is governed by the Lindblad master equation:

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}(t), \hat{\rho}] + \mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (47)$$

7.2 Dissipator

The dissipator has three channels:

$$\begin{aligned} \mathcal{L}_{\text{dec}}[\hat{\rho}] = & \gamma_{\phi} (\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z - \hat{\rho}) \\ & + \gamma_a (n_{\text{th}} + 1) \left(\hat{a} \hat{\rho} \hat{a}^{\dagger} - \frac{1}{2} \{ \hat{a}^{\dagger} \hat{a}, \hat{\rho} \} \right) \\ & + \gamma_a n_{\text{th}} \left(\hat{a}^{\dagger} \hat{\rho} \hat{a} - \frac{1}{2} \{ \hat{a} \hat{a}^{\dagger}, \hat{\rho} \} \right). \end{aligned} \quad (48)$$

Channel	Operator	Rate	Physical meaning
Phase damping	$\hat{\sigma}_z$	γ_ϕ	Loss of analytic coherence
Amplitude damping	$\hat{a}$	$\gamma_a(n_{\text{th}} + 1)$	Substrate relaxation
Thermal pumping	$\hat{a}^\dagger$	$\gamma_a n_{\text{th}}$	Substrate excitation

7.3 Physical interpretation

7.4 Trace preservation

The dissipator is trace-preserving:

$$\text{Tr}(\mathcal{L}_{\text{dec}}[\hat{\rho}]) = 0. \quad (49)$$

8 Reduced dynamics

8.1 Partial traces

The reduced density matrices are

$$\hat{\rho}_1 = \text{Tr}_0(\hat{\rho}) = \sum_{n=0}^{N_0-1} \langle n|_0 \hat{\rho} |n\rangle_0, \quad (50)$$

$$\hat{\rho}_0 = \text{Tr}_1(\hat{\rho}) = \sum_{s \in \{0,1\}} \langle s|_1 \hat{\rho} |s\rangle_1. \quad (51)$$

8.2 Exact reduced equations

The exact evolution equations are

$$\frac{d\hat{\rho}_1}{dt} = -i[\hat{H}_1, \hat{\rho}_1] - i\text{Tr}_0[\hat{H}_{\text{int}}, \hat{\rho}] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1], \quad (52)$$

$$\frac{d\hat{\rho}_0}{dt} = -i[\hat{H}_0, \hat{\rho}_0] - i\text{Tr}_1[\hat{H}_{\text{int}}, \hat{\rho}] + \mathcal{L}_{\text{dec}}^{(0)}[\hat{\rho}_0]. \quad (53)$$

The interaction terms involve correlations and do not close.

9 Observables

9.1 Analytic-mode population

$$\boxed{\langle \hat{\sigma}_z \rangle(t) = \text{Tr}(\hat{\rho}(t) \hat{\sigma}_z)}. \quad (54)$$

Limits:

$$\langle \hat{\sigma}_z \rangle = +1 \iff \text{analytic mode active}, \quad \langle \hat{\sigma}_z \rangle = -1 \iff \text{analytic mode inactive}. \quad (55)$$

9.2 Substrate population

$$\boxed{\langle \hat{n}_0 \rangle(t) = \text{Tr}(\hat{\rho}(t) \hat{a}^\dagger \hat{a})}. \quad (56)$$

9.3 Purity

$$\boxed{\mathcal{P}(t) = \text{Tr}(\hat{\rho}(t)^2)}. \quad (57)$$

9.4 von Neumann entropy

$$S_{vN}(t) = -\text{Tr}(\hat{\rho}(t) \ln \hat{\rho}(t)) = -\sum_i \lambda_i(t) \ln \lambda_i(t). \quad (58)$$

9.5 Steady-state averages

For a given parameter set, we run until $t_{\text{max}} = N_{\text{Rabi}} \cdot T_R$ and average over the final 10% of the trajectory:

$$\langle \hat{O} \rangle_{ss} = \frac{1}{N_{\text{avg}}} \sum_{i \in \text{final window}} \langle \hat{O} \rangle(t_i). \quad (59)$$

10 Control parameters

10.1 Resonance parameter

$$\alpha \equiv \frac{g}{\delta}. \quad (60)$$

Regimes:

- $\alpha \ll 1$: dispersive regime.
- $\alpha \sim 1$: resonant regime.
- $\alpha \gg 1$: ultra-strong coupling.

10.2 Dissipation ratio

$$\mathcal{R} \equiv \frac{g}{\gamma_\phi + \gamma_a}. \quad (61)$$

10.3 Combined parameter space

$$\frac{\mathcal{R}}{\alpha} = \frac{\delta}{\gamma_\phi + \gamma_a} \equiv \mathcal{D}. \quad (62)$$

The phase diagram is effectively two-dimensional in $(\alpha, \mathcal{R})$.

11 Information-theoretic measures

11.1 Channel capacity

$$C = B \log_2(1 + \kappa), \quad \kappa = \frac{g}{\gamma_\phi + \gamma_a} = \mathcal{R}. \quad (63)$$

11.2 Phase distribution

$$P(\Delta\phi) = \frac{e^{\kappa \cos(\Delta\phi - \phi_0)}}{2\pi I_0(\kappa)}. \quad (64)$$

11.3 Phase entropy

$$S_\phi(\kappa) = \ln(2\pi I_0(\kappa)) - \kappa \frac{I_1(\kappa)}{I_0(\kappa)}. \quad (65)$$

Relative entropy:

$$D_{KL}(P\|U) = \ln(2\pi) - S_\phi(\kappa) \geq 0. \quad (66)$$

11.4 Order parameter

$$S(t) = \left| \langle e^{i\Delta\phi(t)} \rangle \right|. \quad (67)$$

12 Phase dynamics

12.1 Kuramoto–Adler equation

$$\frac{d\Delta\phi}{dt} = \Delta\omega - K \sin(\Delta\phi) + \xi(t), \quad (68)$$

with $\Delta\omega = \delta$ and $\xi(t)$ white noise.

12.2 Locking condition

Phase-locking requires

$$K > |\Delta\omega| \iff g > \delta \iff \alpha > 1. \quad (69)$$

13 Numerical scheme

13.1 Vectorization

Reshape $\hat{\rho}$ into a column vector:

$$\boldsymbol{\rho} = \text{vec}(\hat{\rho}) \in \mathbb{C}^{d^2}. \quad (70)$$

The master equation becomes

$$\frac{d\boldsymbol{\rho}}{dt} = \mathcal{L}(t)\boldsymbol{\rho}, \quad (71)$$

with the Liouvillian

$$\mathcal{L}(t) = -i \left(\hat{H}(t) \otimes \hat{I} - \hat{I} \otimes \hat{H}(t)^T \right) + \mathcal{L}_{\text{dec}}. \quad (72)$$

13.2 Implicit integration

We use BDF via `scipy.integrate.solve_ivp` with `rtol` = 10^{-7} , `atol` = 10^{-9} .

13.3 Why BDF?

Stiffness ratio:

$$\frac{\lambda_{\max}}{\lambda_{\min}} \sim \frac{g\sqrt{N_0}}{\gamma_\phi}. \quad (73)$$

For $g = 50$ Hz, $N_0 = 10$, $\gamma_\phi = 0.05$ Hz: ratio $\sim 3 \times 10^4$. Explicit methods are unstable.

13.4 Hermiticity restoration

After each step:

$$\hat{\rho} \leftarrow \frac{1}{2} \left(\hat{\rho} + \hat{\rho}^\dagger \right), \quad \hat{\rho} \leftarrow \frac{\hat{\rho}}{\text{Tr}(\hat{\rho})}. \quad (74)$$

14 Mean-field derivation (full)

This section provides the complete derivation of the mean-field closure used in the reduced dynamics.

14.1 Starting point: full master equation

We start from

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \mathcal{L}_{\text{dec}}[\hat{\rho}], \quad (75)$$

with

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_{\text{int}}, \quad \hat{H}_{\text{int}} = g \left(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+ \right). \quad (76)$$

14.2 Partial trace over the substrate

Taking Tr_0 of both sides:

$$\frac{d\hat{\rho}_1}{dt} = -i\text{Tr}_0[\hat{H}, \hat{\rho}] + \text{Tr}_0\mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (77)$$

Let us evaluate the commutator term by term.

Free analytic term.

$$\text{Tr}_0[\hat{H}_1, \hat{\rho}] = [\hat{H}_1, \text{Tr}_0\hat{\rho}] = [\hat{H}_1, \hat{\rho}_1]. \quad (78)$$

Free substrate term.

$$\text{Tr}_0[\hat{H}_0, \hat{\rho}] = \text{Tr}_0 \left(\hat{H}_0 \hat{\rho} - \hat{\rho} \hat{H}_0 \right). \quad (79)$$

Since $\hat{H}_0$ acts only on $\mathcal{H}_0$, we can write

$$\text{Tr}_0 \left(\hat{H}_0 \hat{\rho} \right) = \sum_n \langle n|_0 \hat{H}_0 \hat{\rho} |n\rangle_0 = \sum_n \omega_0 n \langle n|_0 \hat{\rho} |n\rangle_0, \quad (80)$$

which is not expressible solely in terms of $\hat{\rho}_1$. It contains information about the substrate population distribution.

Interaction term.

$$\text{Tr}_0[\hat{H}_{\text{int}}, \hat{\rho}] = g \left(\text{Tr}_0[\hat{a}^\dagger \hat{\sigma}_- \hat{\rho}] + \text{Tr}_0[\hat{a} \hat{\sigma}_+ \hat{\rho}] \right) - g \left(\text{Tr}_0[\hat{\rho} \hat{a}^\dagger \hat{\sigma}_-] + \text{Tr}_0[\hat{\rho} \hat{a} \hat{\sigma}_+] \right). \quad (81)$$

Using

$$\text{Tr}_0[\hat{a}^\dagger \hat{\sigma}_- \hat{\rho}] = \hat{\sigma}_- \text{Tr}_0[\hat{a}^\dagger \hat{\rho}] = \hat{\sigma}_- \langle \hat{a}^\dagger \hat{\rho} \rangle_0, \quad (82)$$

where we have defined

$$\langle \hat{a}^\dagger \hat{\rho} \rangle_0 \equiv \text{Tr}_0[\hat{a}^\dagger \hat{\rho}]. \quad (83)$$

This is a *correlation* between the substrate operator $\hat{a}^\dagger$ and the joint state $\hat{\rho}$.

14.3 The correlation problem

Define the two-sector correlations:

$$C_{a^\dagger\sigma_-} \equiv \langle \hat{a}^\dagger \hat{\sigma}_- \rangle - \langle \hat{a}^\dagger \rangle \langle \hat{\sigma}_- \rangle, \quad (84)$$

$$C_{a\sigma_+} \equiv \langle \hat{a} \hat{\sigma}_+ \rangle - \langle \hat{a} \rangle \langle \hat{\sigma}_+ \rangle. \quad (85)$$

Then the reduced equation becomes

$$\frac{d\hat{\rho}_1}{dt} = -i[\hat{H}_1 + \hat{H}_{\text{int}}^{(1)}, \hat{\rho}_1] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1] + \mathcal{C}^{(1)}, \quad (86)$$

where

$$\hat{H}_{\text{int}}^{(1)} = g \left(\langle \hat{a}^\dagger \rangle_{\hat{\rho}_0} \hat{\sigma}_- + \langle \hat{a} \rangle_{\hat{\rho}_0} \hat{\sigma}_+ \right), \quad (87)$$

and $\mathcal{C}^{(1)}$ contains the correlation terms

$$\mathcal{C}^{(1)} = -ig \left(C_{a^\dagger\sigma_-} [\hat{\sigma}_-, \cdot] + C_{a\sigma_+} [\hat{\sigma}_+, \cdot] \right). \quad (88)$$

14.4 Hartree–Fock factorization

We now invoke the Hartree–Fock approximation:

$$\boxed{\hat{\rho} \approx \hat{\rho}_0 \otimes \hat{\rho}_1.} \quad (89)$$

Under this approximation:

$$\langle \hat{a}^\dagger \hat{\sigma}_- \rangle \approx \langle \hat{a}^\dagger \rangle_{\hat{\rho}_0} \langle \hat{\sigma}_- \rangle_{\hat{\rho}_1}, \quad (90)$$

$$\langle \hat{a} \hat{\sigma}_+ \rangle \approx \langle \hat{a} \rangle_{\hat{\rho}_0} \langle \hat{\sigma}_+ \rangle_{\hat{\rho}_1}. \quad (91)$$

Therefore the correlations vanish:

$$C_{a^\dagger\sigma_-} \rightarrow 0, \quad C_{a\sigma_+} \rightarrow 0. \quad (92)$$

14.5 Closed system

Under Hartree–Fock, we obtain the closed pair:

$$\boxed{\frac{d\hat{\rho}_1}{dt} = -i[\hat{H}_1 + \hat{H}_{\text{int}}^{(1)}(\hat{\rho}_0), \hat{\rho}_1] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1]} \quad (93)$$

$$\boxed{\frac{d\hat{\rho}_0}{dt} = -i[\hat{H}_0 + \hat{H}_{\text{int}}^{(0)}(\hat{\rho}_1), \hat{\rho}_0] + \mathcal{L}_{\text{dec}}^{(0)}[\hat{\rho}_0]} \quad (94)$$

with

$$\hat{H}_{\text{int}}^{(1)}(\hat{\rho}_0) = g \left(\langle \hat{a}^\dagger \rangle_{\hat{\rho}_0} \hat{\sigma}_- + \langle \hat{a} \rangle_{\hat{\rho}_0} \hat{\sigma}_+ \right), \quad (95)$$

$$\hat{H}_{\text{int}}^{(0)}(\hat{\rho}_1) = g \left(\langle \hat{\sigma}_- \rangle_{\hat{\rho}_1} \hat{a}^\dagger + \langle \hat{\sigma}_+ \rangle_{\hat{\rho}_1} \hat{a} \right). \quad (96)$$

14.6 Validity conditions

The Hartree–Fock approximation is valid when

$$\left| C_{a^\dagger \sigma_-} \right| \ll \left| \langle \hat{a}^\dagger \rangle \langle \hat{\sigma}_- \rangle \right|. \quad (97)$$

This is satisfied when:

1. The number of substrate quanta is large: $\langle \hat{n}_0 \rangle \gg 1$ (classical limit of the oscillator).
2. The analytic mode is weakly entangled with the substrate: $S_{vN}(\hat{\rho}_1) \ll \ln 2$.
3. The coupling is small compared to the detuning: $\alpha \ll 1$.

For the numerical parameters of Track A, condition (3) is satisfied only in the dispersive regime. In the resonant and strong-coupling regimes, the mean-field approximation breaks down, and the full master equation must be integrated. This is why the numerical simulations use the full Liouvillian, not the mean-field closure.

Remark 1. The mean-field derivation is included for pedagogical completeness. The numerical results of the main paper do not rely on it.

15 Density matrix expansion in the Jaynes–Cummings eigenbasis

This section provides the expansion of the density matrix in the eigenbasis of the Jaynes–Cummings Hamiltonian, which is the natural basis for understanding the dynamics.

15.1 Dressed-state basis

For each excitation manifold $N \in \{1, 2, \dots, N_0\}$, define

$$|+, N\rangle = \cos \theta_N |N, 0\rangle + \sin \theta_N |N-1, 1\rangle, \quad (98)$$

$$|-, N\rangle = -\sin \theta_N |N, 0\rangle + \cos \theta_N |N-1, 1\rangle. \quad (99)$$

The vacuum state $|0, 0\rangle$ is a singlet and is not part of any two-dimensional manifold.

The dressed-state basis is

$$\mathcal{B}_{JC} = \left\{ |0, 0\rangle, \bigcup_{N=1}^{N_0} \{|+, N\rangle, |-, N\rangle\} \right\}. \quad (100)$$

Its dimension is $1 + 2N_0$, which exceeds $\dim \mathcal{H} = 2N_0$ by one. The overcounting is resolved by noting that the highest manifold $N = N_0$ has only one state within the truncated space. For the numerical implementation, we use the product basis $\{|n, s\rangle\}$ directly, and the dressed states are used only for interpretation.

15.2 Hamiltonian in the dressed basis

In the dressed basis, the free Hamiltonian is diagonal:

$$\hat{H}_0 |\pm, N\rangle = E_{\pm}^{(N)} |\pm, N\rangle, \quad (101)$$

with

$$E_{\pm}^{(N)} = \omega_0(N-1) + \omega_1/2 + \delta/2 \pm \frac{1}{2}\Omega_R^{(N)}. \quad (102)$$

The interaction $\hat{H}_{\text{int}}$ is also diagonal in the dressed basis at fixed N , but off-diagonal between manifolds:

$$\langle \pm, N | \hat{H}_{\text{int}} | \pm', N' \rangle = g \left(\sqrt{N} \cos \theta_N \sin \theta_{N-1} \delta_{N', N-1} + \sqrt{N+1} \sin \theta_N \cos \theta_{N+1} \delta_{N', N+1} \right) \delta_{\pm, \pm'}. \quad (103)$$

15.3 Density matrix in the dressed basis

Write the density matrix as

$$\hat{\rho} = \sum_{N,N'} \sum_{\sigma,\sigma' \in \{+,-\}} \rho_{\sigma,\sigma'}^{(N,N')} |\sigma, N\rangle \langle \sigma', N'| + \text{vacuum terms.} \quad (104)$$

The coefficients are

$$\rho_{\sigma,\sigma'}^{(N,N')} = \langle \sigma, N | \hat{\rho} | \sigma', N' \rangle. \quad (105)$$

For a thermal state at temperature T :

$$\rho_{\sigma,\sigma'}^{(N,N')} \propto e^{-\beta E_{\sigma}^{(N)}} \delta_{N,N'} \delta_{\sigma,\sigma'}. \quad (106)$$

15.4 Master equation in the dressed basis

The master equation becomes

$$\frac{d\rho_{\sigma,\sigma'}^{(N,N')}}{dt} = -i \left(E_{\sigma}^{(N)} - E_{\sigma'}^{(N')} \right) \rho_{\sigma,\sigma'}^{(N,N')} + \sum_{\tau,\tau',M,M'} \mathcal{L}_{\sigma\sigma'\tau\tau'}^{(N,N';M,M')} \rho_{\tau,\tau'}^{(M,M')}. \quad (107)$$

The coherent part is diagonal in (N, N') and (σ, σ') for the free evolution. The dissipative part couples different manifolds.

15.5 Secular approximation

In the weak-dissipation limit, the off-diagonal elements oscillate rapidly at frequencies $|E_{\sigma}^{(N)} - E_{\sigma'}^{(N')}|$. When these frequencies are much larger than the dissipation rates, we can average over the fast oscillations (secular approximation):

$$\frac{d\rho_{\sigma,\sigma'}^{(N,N')}}{dt} \approx -i \left(E_{\sigma}^{(N)} - E_{\sigma'}^{(N')} \right) \rho_{\sigma,\sigma'}^{(N,N')} + \delta_{N,N'} \delta_{\sigma,\sigma'} \sum_{\tau,M} \Gamma_{\sigma\sigma,\tau\tau}^{(N,N;M,M)} \rho_{\tau,\tau'}^{(M,M)}. \quad (108)$$

This reduces the master equation to a Pauli-type rate equation for populations, valid in the weak-coupling regime.

15.6 Physical interpretation

The dressed-state picture provides a clear interpretation of the dynamics:

- At fixed N , the states $|+, N\rangle$ and $|- , N\rangle$ are the eigenstates of the coupled system. Their populations are the physically relevant quantities.
- The decay of the excited dressed state to lower manifolds corresponds to the emission of substrate quanta.
- The coupling g determines the mixing between the bare states $|N, 0\rangle$ and $|N-1, 1\rangle$. When $g = 0$, the bare states are eigenstates. When $g \rightarrow \infty$, the dressed states become equal superpositions.

The numerical results of Track A can be reinterpreted as follows:

- The steady-state value $\langle \hat{\sigma}_z \rangle_{ss}$ is a weighted sum of dressed-state populations.
- The substrate excitation $\langle \hat{n}_0 \rangle_{ss}$ is the total population of manifolds with $N \geq 1$.
- The boundary $\alpha_{\text{cruce}}(\gamma_{\phi})$ corresponds to a change in the dominant dressed-state manifold.

16 Linear stability analysis of the Liouvillian

This section provides the linear stability analysis of the Liouvillian superoperator, which determines the approach to the steady state.

16.1 Vectorization of the master equation

Define the vectorization of the density matrix:

$$\boldsymbol{\rho} = \text{vec}(\hat{\rho}) \in \mathbb{C}^{d^2}. \quad (109)$$

The master equation becomes

$$\frac{d\boldsymbol{\rho}}{dt} = \mathcal{L}(t)\boldsymbol{\rho}, \quad (110)$$

with the Liouvillian

$$\mathcal{L}(t) = -i \left(\hat{H}(t) \otimes \hat{I} - \hat{I} \otimes \hat{H}(t)^T \right) + \mathcal{L}_{\text{dec}}, \quad (111)$$

where

$$\mathcal{L}_{\text{dec}} = \mathcal{L}_{\text{phase}} + \mathcal{L}_{\text{amp}} + \mathcal{L}_{\text{therm}}. \quad (112)$$

For the phase-damping channel:

$$\mathcal{L}_{\text{phase}} = \gamma_{\phi} \left(\hat{\sigma}_z \otimes \hat{\sigma}_z^T - \hat{I} \otimes \hat{I} \right). \quad (113)$$

For the amplitude-damping channel:

$$\mathcal{L}_{\text{amp}} = \gamma_a(n_{\text{th}} + 1) \left(\hat{a} \otimes \hat{a}^* - \frac{1}{2} \hat{a}^\dagger \hat{a} \otimes \hat{I} - \frac{1}{2} \hat{I} \otimes (\hat{a}^\dagger \hat{a})^T \right). \quad (114)$$

For the thermal-pumping channel:

$$\mathcal{L}_{\text{therm}} = \gamma_a n_{\text{th}} \left(\hat{a}^\dagger \otimes (\hat{a}^\dagger)^* - \frac{1}{2} \hat{a} \hat{a}^\dagger \otimes \hat{I} - \frac{1}{2} \hat{I} \otimes (\hat{a} \hat{a}^\dagger)^T \right). \quad (115)$$

16.2 Steady state

For constant g (time-independent Liouvillian), the steady state $\boldsymbol{\rho}_{ss}$ satisfies

$$\mathcal{L}\boldsymbol{\rho}_{ss} = 0. \quad (116)$$

This is the null eigenvector of the Liouvillian.

16.3 Spectral decomposition

The Liouvillian is a linear operator on $\mathbb{C}^{d^2}$. Its eigenvalues $\{\lambda_\mu\}_{\mu=0}^{d^2-1}$ satisfy

$$\mathcal{L}\boldsymbol{\rho}_\mu = \lambda_\mu \boldsymbol{\rho}_\mu. \quad (117)$$

Trace preservation requires that one eigenvalue is zero:

$$\lambda_0 = 0, \quad \boldsymbol{\rho}_0 = \boldsymbol{\rho}_{ss}. \quad (118)$$

All other eigenvalues have negative real parts:

$$\text{Re}(\lambda_\mu) < 0 \quad \forall \mu \geq 1. \quad (119)$$

16.4 Relaxation rates

The relaxation rate to the steady state is set by the spectral gap:

$$\Delta \equiv -\max_{\mu \geq 1} \operatorname{Re}(\lambda_\mu) = |\operatorname{Re}(\lambda_1)|. \quad (120)$$

This is the slowest decay rate. The corresponding time scale is

$$\tau_{\text{relax}} = \frac{1}{\Delta}. \quad (121)$$

16.5 Structure of the spectrum

The spectrum of the Liouvillian has the following structure:

Dark states. The steady state ρ_{ss} has $\lambda_0 = 0$. Other dark states may exist for special parameter values.

Coherent modes. The eigenmodes corresponding to oscillations at the Rabi frequency have

$$\lambda_\mu = -\frac{\gamma_{\text{eff}}}{2} \pm i\Omega_R^{(N)}. \quad (122)$$

Their real part is the effective decay rate, and their imaginary part is the Rabi frequency.

Dissipative modes. The eigenmodes corresponding to purely dissipative processes have purely real (negative) eigenvalues:

$$\lambda_\mu = -\gamma_\mu, \quad \gamma_\mu > 0. \quad (123)$$

These are the classical relaxation modes.

16.6 Effective decay rate

For the Jaynes–Cummings model with dissipation, the effective decay rate of the coherence is

$$\gamma_{\text{eff}} = \frac{\gamma_\phi + \gamma_a \langle \hat{n}_0 \rangle}{2}. \quad (124)$$

This combines phase damping and the population-dependent amplitude damping.

16.7 Overdamped vs underdamped

The dynamics are:

- **Underdamped** if $\Omega_R > \gamma_{\text{eff}}/2$: oscillations visible.
- **Critical** if $\Omega_R = \gamma_{\text{eff}}/2$: fastest return to steady state.
- **Overdamped** if $\Omega_R < \gamma_{\text{eff}}/2$: no oscillations.

The boundary between regimes is

$$\Omega_R = \frac{\gamma_{\text{eff}}}{2} \iff \sqrt{\delta^2 + 4g^2} = \frac{\gamma_\phi + \gamma_a \langle \hat{n}_0 \rangle}{4}. \quad (125)$$

16.8 Comparison with numerical results

For the parameters of Track A:

- Endogenous ($g = 5$, $\gamma_\phi = 0.40$): $\Omega_R = 34.7$ Hz, $\gamma_{\text{eff}} = 0.21$ Hz. Underdamped.
- Mixed ($g = 17$, $\gamma_\phi = 1.36$): $\Omega_R = 48.1$ Hz, $\gamma_{\text{eff}} = 0.70$ Hz. Underdamped.
- Exogenous ($g = 50$, $\gamma_\phi = 4.90$): $\Omega_R = 104.0$ Hz, $\gamma_{\text{eff}} = 2.46$ Hz. Underdamped.

All three regimes are underdamped. The observed difference in steady-state substrate excitation is not due to the damping regime but due to the different equilibrium populations.

16.9 Time-averaged Liouvillian

For time-dependent $g(t)$, the Liouvillian is time-dependent. The relevant quantity for the long-time behavior is the time-averaged Liouvillian:

$$\bar{\mathcal{L}} = \frac{1}{T} \int_0^T \mathcal{L}(t) dt. \quad (126)$$

For the DMN-modulated coupling:

$$\bar{\mathcal{L}} = \mathcal{L}_0 + \bar{g} \left(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+ \right) \otimes \hat{I} + \text{h.c.}, \quad (127)$$

with

$$\bar{g} = \frac{1}{T} \int_0^T g(t) dt. \quad (128)$$

16.10 Adiabatic approximation

When $\tau_{\text{DMN}} \gg T_R$, the DMN evolves slowly compared to the Rabi dynamics. In this regime, the adiabatic approximation applies: the system follows the instantaneous steady state of $\mathcal{L}(t)$. The long-time average is then

$$\langle \hat{O} \rangle_{ss} \approx \frac{1}{T} \int_0^T \langle \hat{O} \rangle_{ss}[g(t)] dt. \quad (129)$$

This justifies the numerical procedure of averaging over the final 10% of the trajectory.

16.11 Spectral gap and the phase diagram

The boundary between analytic-dominated and substrate-dominated regimes corresponds to a change in the dominant eigenmode of the Liouvillian:

- **Analytic-dominated:** the dominant slow mode has $\langle \hat{\sigma}_z \rangle_{ss} > 0$.
- **Substrate-dominated:** the dominant slow mode has $\langle \hat{\sigma}_z \rangle_{ss} < 0$.

The boundary $\alpha_{\text{cruce}}(\gamma_\phi)$ is the locus in parameter space where the two slow modes exchange dominance.

Remark 2. For finite N_0 , the Liouvillian spectrum is discrete. The phase boundary is sharp only in the limit $N_0 \rightarrow \infty$. For finite N_0 , it is a crossover rather than a true phase transition. The numerical results of Track A confirm this: the transition is smooth, not sharp.

16.12 Summary of the stability analysis

The linear stability analysis provides:

1. A rigorous definition of the steady state as the null eigenvector of the Liouvillian.
2. A characterization of the relaxation rate via the spectral gap.
3. A criterion for underdamped vs overdamped dynamics.
4. A framework for understanding the phase boundary as a change in dominant eigenmode.

These results underpin the numerical findings of the main paper and provide the theoretical foundation for the falsifiable predictions.

17 Parameter table

Table 1: Complete list of parameters used in Track A.

Symbol	Description	Value	Status
ω_0	Substrate frequency	6 Hz	free
ω_1	Analytic frequency	40 Hz	free
δ	Detuning	34 Hz	derived
N_0	Substrate truncation	10 (scan: 6)	numerical
g_0	Maximal coupling	20 Hz	free
λ	DMN sigmoid stiffness	8	free
A_{th}	DMN threshold	0.5	free
A_0	DMN baseline	0.3	free
τ_{DMN}	DMN relaxation time	2 s	free
σ	DMN noise amplitude	0.05	free
γ_ϕ	Phase damping rate	0.5 Hz	free
γ_a	Amplitude damping rate	0.1 Hz	free
n_{th}	Thermal occupation	0.05	free

18 Summary of key equations

Master equation.

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}(t), \hat{\rho}] + \mathcal{L}_{\text{dec}}[\hat{\rho}]. \quad (130)$$

Hamiltonian.

$$\hat{H}(t) = \omega_0 \hat{a}^\dagger \hat{a} + \omega_1 \hat{\sigma}_+ \hat{\sigma}_- + g(t) \left(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+ \right). \quad (131)$$

Dissipator.

$$\mathcal{L}_{\text{dec}}[\hat{\rho}] = \gamma_\phi (\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z - \hat{\rho}) + \gamma_a (n_{\text{th}} + 1) \mathcal{D}[\hat{a}] \hat{\rho} + \gamma_a n_{\text{th}} \mathcal{D}[\hat{a}^\dagger] \hat{\rho}, \quad (132)$$

with $\mathcal{D}[\hat{L}] \hat{\rho} = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \{ \hat{L}^\dagger \hat{L}, \hat{\rho} \}$.

Coupling.

$$g(t) = g_0 \eta(A_{\text{DMN}}(t)), \quad \eta(A) = \frac{1}{1 + e^{\lambda(A - A_{\text{th}})}}. \quad (133)$$

Control parameters.

$$\alpha = \frac{g}{\delta}, \quad \mathcal{R} = \frac{g}{\gamma_\phi + \gamma_a}. \quad (134)$$

Observables.

$$\langle \hat{\sigma}_z \rangle = \text{Tr}(\hat{\rho} \hat{\sigma}_z), \quad \langle \hat{n}_0 \rangle = \text{Tr}(\hat{\rho} \hat{a}^\dagger \hat{a}). \quad (135)$$

Mean-field closure.

$$\frac{d\hat{\rho}_1}{dt} = -i[\hat{H}_1 + \hat{H}_{\text{int}}^{(1)}(\hat{\rho}_0), \hat{\rho}_1] + \mathcal{L}_{\text{dec}}^{(1)}[\hat{\rho}_1]. \quad (136)$$

Dressed states.

$$|\pm, N\rangle = \pm \sin \theta_N |N, 0\rangle + \cos \theta_N |N-1, 1\rangle, \quad \theta_N = \frac{1}{2} \arctan \left(\frac{2g\sqrt{N}}{\delta} \right). \quad (137)$$

Liouvillian.

$$\mathcal{L}(t) = -i \left(\hat{H}(t) \otimes \hat{I} - \hat{I} \otimes \hat{H}(t)^T \right) + \mathcal{L}_{\text{dec}}. \quad (138)$$

Spectral gap.

$$\Delta = -\max_{\mu \geq 1} \text{Re}(\lambda_\mu). \quad (139)$$

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